

**Mid-Term-1 Examination – 2025**

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, SEPTEMBER 2025

**Paper Code: ES-201****Subject: Computational Methods****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from the remaining.**

| Q. No.          | Question                                                                                                                                                                                                                                                                                             | Marks | CO    |       |       |       |       |                 |    |    |    |    |    |     |     |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|-------|-------|-----------------|----|----|----|----|----|-----|-----|
| 1(a)            | Write Rolle's theorem and apply it to the function $f(x) = \sin x / e^x$ in $(0, \pi)$ .                                                                                                                                                                                                             | [2.5] | CO1   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 1(b)            | Evaluate $\Delta^3 [(1 - x)(1 - 2x)(1 - 3x)]$ , the interval of differencing being unity.                                                                                                                                                                                                            | [2]   | CO2   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 1(c)            | Evaluate $\sqrt[3]{24}$ using the Newton-Raphson method.                                                                                                                                                                                                                                             | [2.5] | CO1   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 1(d)            | Evaluate $\int_0^6 [1 / (1 + x^2)] dx$ by using Simpson's 1/3 rule.                                                                                                                                                                                                                                  | [3]   | CO2   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 2(a)            | Find the root of the equation $x^3 - 2x - 5 = 0$ using the secant method, correct to three decimal places.                                                                                                                                                                                           | [5]   | CO1   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 2(b)            | Find the minimum value of $f(x) = x^2 + 54/x$ within the interval $[0, 5]$ using the Fibonacci search method ( $n = 3$ ).                                                                                                                                                                            | [5]   | CO1   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 3(a)            | Find the minimum value of $f(x) = 0.25x^4 - x^2 - 5x + 1$ in the interval $[0, 3]$ using Newton method given $\varepsilon = 0.01$ .                                                                                                                                                                  | [5]   | CO1   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 3(b)            | From the following table, estimate the number of students who obtained marks between 40 and 45: <table><tr><td>Marks</td><td>30–40</td><td>40–50</td><td>50–60</td><td>60–70</td><td>70–80</td></tr><tr><td>No. of students</td><td>31</td><td>42</td><td>51</td><td>35</td><td>31</td></tr></table> | Marks | 30–40 | 40–50 | 50–60 | 60–70 | 70–80 | No. of students | 31 | 42 | 51 | 35 | 31 | [5] | CO2 |
| Marks           | 30–40                                                                                                                                                                                                                                                                                                | 40–50 | 50–60 | 60–70 | 70–80 |       |       |                 |    |    |    |    |    |     |     |
| No. of students | 31                                                                                                                                                                                                                                                                                                   | 42    | 51    | 35    | 31    |       |       |                 |    |    |    |    |    |     |     |
| 4(a)            | Using Lagrange's formula, express the function $(3x^2 + x + 1) / [(x - 1)(x - 2)(x - 3)]$ as a sum of partial fractions.                                                                                                                                                                             | [5]   | CO2   |       |       |       |       |                 |    |    |    |    |    |     |     |
| 4(b)            | Evaluate $\int_0^1 [dx / (1 + x)]$ correct to three decimal places using Romberg's method.                                                                                                                                                                                                           | [5]   | CO2   |       |       |       |       |                 |    |    |    |    |    |     |     |

\*\*\*\* End of Question Paper \*\*\*\*